Selected-Frame Spatial Structure in Clustered Heritage Data: A Non-Confirmatory Worked Example

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Abstract

Spatial analyses of heritage data routinely impose coordinate-dependent scoring functions, yet the literature seldom examines how the choice of reference frame interacts with the geographic clustering typical of cultural corpora. This paper reports a weak selected-frame signal in the UNESCO Cultural and Mixed World Heritage corpus ($N = 1011$): a coordinate-dependent sweep selects a frame near 3° spacing and the ∼35°E corridor, within which 132/1011 sites fall in Tier-A+ (13.1% versus the 10% geometric null). Conditional diagnostics within that frame are positive, and related same-footprint stability appears in Wikidata Q180987 stupas ($N = 229$) and the OWTRAD Silk Road network ($N = 1674$).

The same evidence does not validate an invariant-period interpretation. The full-corpus Rayleigh test on raw longitudes is null ($p_{\text{perm}} = 0.3716$), the Americas subcorpus does not support extension outside the Eurasian footprint, and the Wikidata and OWTRAD checks are geographically confounded because all three corpora are concentrated along the same broad corridor. Wikidata is also partly in-sample for the stability check because it is used in tier-threshold derivation. Shared geography plus frame selection is therefore a sufficient alternative explanation for the observed same-frame behaviour.

The result is non-confirmatory rather than null in every respect. The analysis locates the signal at the selected-frame level: something measurable is present under the chosen representation, but current tests do not show that it is independent, invariant, or portable beyond the Eurasian footprint. The methodological contribution is a seven-stage audit pipeline (unit discovery, anchor characterisation, tier-threshold derivation, within-frame diagnostics, invariant boundary tests, geographic-concentration nulls, and cross-corpus stability checks) for preventing

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weak selected-frame spatial signals from being overinterpreted. All code, data, and reproduction scripts are deposited at Zenodo.

Keywords: spatial statistics, frame dependence, cultural analytics, UNESCO World Heritage, non-confirmatory analysis, shared-geography confounding, researcher degrees of freedom, diagnostic audit

1 Introduction

Spatial analysis of heritage data depends on choices that are rarely made explicit: the coordinate system, the reference frame, the scoring or distance function, and the scale at which proximity is evaluated. When a corpus is geographically clustered — as most cultural heritage datasets are — these choices interact with the clustering geometry. The interaction can produce weak but statistically detectable *within-frame* patterns whose status remains unresolved under more invariant tests. More subtly, when the same frame is applied to several corpora that share geographic footprint, the resulting enrichment can appear stable across corpora. Such stability is an empirical observation, but it is not automatically independent replication: common geography viewed through the same representational choice can produce the same behaviour.

This paper is concerned with that intermediate status. Cross-corpus testing at frozen parameters has been proposed as a remedy for the well-known flexibility of single-corpus spatial analysis (Thom, 1967; Kendall, 1974; Freeman, 1976). The remedy is partial: agreement across geographically overlapping corpora is not equivalent to agreement across statistically independent samples of the underlying spatial process. Without an explicit invariant comparison and without out-of-footprint support, a selected-frame signal can remain real as a within-representation pattern while still failing to become confirmatory evidence for spatial periodicity.

Prior work on spatial periodicity in monument distributions (Thom, 1967; Kendall, 1974; Freeman, 1976) was criticised for investigator-assembled site lists, ad hoc unit choices, and an absence of principled tests separating frame-dependent structure from invariant periodicity. More recent probability-based frameworks for archaeoastronomical alignments (Magli, 2013, 2016; Ruggles, 1999, 2015; Silva & Steele, 2020) and the ICOMOS-IAU thematic study (Ruggles & Cotte, 2010) have advanced the field but have not addressed the specific problem of how weak selected-frame structure should be audited under shared-geography conditions. Adjacent literature on researcher degrees of freedom and the garden of forking paths (Gelman & Loken, 2014) provides a general vocabulary for the flexibility we examine here, but is not heritage-specific.

Contribution. We contribute a reproducible audit pipeline that addresses each of the failure modes above. The pipeline uses externally curated corpora, parameter sweeps with full reporting of unit and anchor selection, tier thresholds derived from a held-out corpus's own dispersion, an explicit invariant boundary-condition test, geographic-concentration nulls, and a cross-corpus stability stage that distinguishes same-footprint checks from out-of-footprint validation. Crucially, the pipeline is structured so that disagreement between within-frame diagnostics and the invariant test *locates* the result at the selected-frame level, rather than collapsing the two statuses into either a positive finding or a complete null. The pipeline is demonstrated on the UNESCO Cultural and Mixed World Heritage corpus ($N = 1011$) with stability checks against the Wikidata Q180987 stupa corpus ($N = 229$) and the OWTRAD attested Silk Road network ($N = 1674$ unique endpoints).

What the worked example shows. The example is informative because it produces a weak selected-frame pattern that is not purely a single-site accident and not simply absent. A parameter sweep selects a frame near 3° spacing and the ∼35°E corridor; under that frame, the UNESCO corpus has 132/1011 Tier-A+ sites (13.1% versus a 10% geometric null), and Stage 4 conditional diagnostics are positive. The same frozen parameters show related behaviour in Wikidata stupas and OWTRAD route nodes. Read in isolation, these results would be easy to overstate. The full-corpus Rayleigh test on raw longitudes, however, is null ($p_{\text{perm}} = 0.3716$); the Americas sub-corpus does not support extension outside the Eurasian footprint; and all three positive corpora are concentrated along the same corridor. The paper therefore reports the result as unresolved and non-confirmatory: a weak selected-frame signal exists, but current tests do not validate it as invariant spatial periodicity.

Paper structure. Section 2 describes the pipeline in general terms. Section 3 describes the corpora. Section 4 presents the worked example, organised by pipeline stage, with the invariant boundary-condition test (Section 4.5) as the main limit on interpretation. Section 5 reports cross-corpus stability under frozen parameters and shows why, in this case, that stability does not warrant an invariant interpretation. Section 6 interprets what the pipeline reveals. Section 7 draws implications for cultural analytics practice. Section 8 concludes.

2 The Pipeline

The pipeline consists of seven stages, described here in general terms and instantiated for the worked example in Sections 4–5. The stages are designed jointly: each later stage interrogates assumptions a naive use of an earlier stage might leave unchallenged.

Stage 1: Unit discovery. A sweep over candidate angular spacings ω identifies which spacing produces the strongest node-proximity enrichment in the primary corpus. A degree-denominated Monte Carlo null tests whether the selected spacing exceeds what geographic clustering alone would produce. The number of spacings tested and the resolution of the sweep are reported as part of the inferential design, not concealed. This stage selects a frame for audit; it does not establish that the selected spacing is an invariant period of the corpus.

Stage 2: Anchor characterisation. For the selected ω, a global sweep over candidate reference longitudes ϕ (at fine resolution across $[0°, 360°\mathrm{E})$) identifies the corridor that maximises within-frame enrichment. A permutation null calibrates whether the maximum is unusual relative to the same sweep over permuted longitudes. The maximum-over-anchors statistic accounts for the search; nominally significant single-anchor p-values that ignore the sweep are not reportable as discovery. Marginal anchor-sweep results are treated as boundary conditions on the selected frame, not as grounds for dismissing the selected-frame signal.

Stage 3: Tier-threshold derivation. Proximity tiers are defined by angular thresholds, ideally derived from a held-out corpus's angular dispersion so that tier widths do not depend on the primary data. This stage fixes the descriptive vocabulary (A++, A+, A, B, C) used by downstream diagnostics. Tier widths derived from a corpus that is later used in cross-corpus stability checks (Stage 7) are partly in-sample for that check, and the limitation is acknowledged where it applies.

Stage 4: Within-frame diagnostics. Conditional on the selected (ω, ϕ) frame, a battery of tests characterises the within-frame structure: a von Mises mixture test on a morphologically defined sub-corpus, a Spearman cluster-asymmetry test, a phase-rotation null, and a Tier-A+ enrichment summary. All reported p-values at this stage are conditional on the selected frame and adjust within the reported family (Holm step-down). They do not generalise to the unconditional question of whether invariant periodicity exists in the data, but they do indicate whether the selected-frame signal has internal structure beyond a single highlighted site.

Stage 5: Boundary-condition tests. At least one *frame-invariant* test — one that does not assume the selected reference frame — checks whether the within-frame structure extends to invariant periodicity. The Rayleigh test on raw longitudes is the canonical choice. Disagreement between Stages 4 and 5 is the main inferential pivot: it locates the result at the selected-frame level rather than validating an invariant-period interpretation. It also prevents the within-frame conditional p-values from being mistaken for unconditional evidence.

Stage 6: Geographic-concentration nulls. Targeted null models test whether the within-frame enrichment in sub-populations exceeds what their longitude footprint alone would produce. A within-subgroup bootstrap (Null A) and a restricted geographic draw (Null C) address this question from complementary directions. An out-of-footprint sub-corpus is examined for whether the pattern extends beyond the geographic concentration of the primary data.

Stage 7: Cross-corpus stability. The selected frame is applied with all parameters frozen to independently assembled corpora. The same diagnostics are run. Two questions are asked: (i) does the frame organise the additional corpora to comparable degree, and (ii) does the same-frame stability survive a comparison to alternative reference longitudes (anchor-specificity check)? Critically, agreement on (i) is not interpreted as confirmation of invariant structure unless the test corpora are out of the primary corpus's geographic footprint *and* Stage 5 is positive. Where the test corpora share footprint with the primary corpus, cross-corpus agreement is reported as same-footprint stability under a shared representation, not independent replication.

3 Data

3.1 Primary Corpus: UNESCO World Heritage

The UNESCO World Heritage XML dataset (UNESCO, 2024) contains 1248 inscribed nominations. Retaining Cultural and Mixed sites (excluding 235 Natural-only) and dropping 2 entries lacking geocoordinates yields $N = 1011$ sites. The corpus over-represents Europe and the Mediterranean-to-South-Asian corridor (Europe and North America: 50%; Asia-Pacific: 23%; Latin America and Caribbean: 11%; Arab States: 9%; Africa: 7%). This geographic concentration is itself a structural feature that interacts with any harmonic scoring scheme; the pipeline characterises rather than assumes away that interaction.

Some UNESCO properties are serial or multi-component nominations; the XML dataset provides a single representative coordinate. Within-frame results therefore partly reflect the coordinate structure of the corpus as published, not exclusively the underlying spatial distribution of the inscribed properties.

3.2 Stability-Check Corpora

Two independently assembled datasets serve as cross-corpus stability checks (Stage 7).

Wikidata Q180987 stupas ($N = 229$). A corpus of stupa sites retrieved from Wikidata (Tenenbaum, 2026). It overlaps geographically and culturally with UNESCO's South and Southeast Asian inscriptions. The Wikidata corpus is also used in Stage 3 to derive tier thresholds; it is therefore partly in-sample for the Stage 7 check, a limitation flagged where it applies.

OWTRAD Silk Road network ($N = 1674$ unique endpoints). The Old World Trade Routes database (Ciolek, 2004), comprising 1946 route segments. This corpus represents historical trade connectivity rather than monumental architecture and is not used in any earlier pipeline stage; it is out-of-sample with respect to parameter derivation, though not out-of-footprint geographically.

All three corpora are concentrated within the Eurasian corridor; the Stage 7 design therefore tests *same-footprint* cross-corpus stability. A genuinely out-of-footprint validation corpus is not available at present, and the implications of that absence are central to the worked example's interpretation (Sections 5.3, 6).

3.3 Domed-Monument Sub-corpus

A keyword filter (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) applied to UNESCO site names and description texts yields $N = 90$ sites. The keyword set was refined iteratively during analysis; results from this sub-corpus are explicitly hypothesis-generating and are reported as exploratory subset characterisation, not as confirmatory inference. A context-validated subset ($N = 83$, 7 ambiguous sites rejected; Appendix C) is used for sensitivity checks.

4 Worked Example: UNESCO Heritage Corpus

4.1 Stage 1: Unit Discovery

A sweep over candidate spacings from 0.5° to 15° identifies a spacing near 3° as the highest-scoring parameter. Adjacent non-harmonic spacings (1.8°, 2.1°, 2.7°, 3.3°, 3.6°) do not reach significance. The 1.5° and 6° spacings also score well, but both are mathematical consequences of the 3° structure (every 3° node is a 1.5° node; 6° nodes are a strict subset), so the sweep is consistent with 3° as the best-scoring spacing in the parameter family tested.

Two interpretive points follow. First, "best-scoring spacing in the sweep" is not equivalent to "preferred spacing of the underlying spatial process"; the unconditional question is addressed at Stage 5. Second, the 3° value is treated throughout as the pipeline's selected analytic spacing, not as a demonstrated external unit.

4.2 Stage 2: Anchor Characterisation

For the selected spacing near 3°, a global sweep over 36,000 trial anchors (0° to 360°E, 0.01° resolution) identifies the ∼35°E longitude corridor as the maximum-enrichment anchor. The maximum places 132 A+ sites at the 99.7$^{\text{th}}$ percentile of the trial-anchor distribution, with a nearby secondary peak (128 A+, 97.7$^{\text{th}}$ percentile) ∼0.04° away. The relevant empirical object is the ∼35°E corridor; the existence of two close anchors with similar scores reflects the corridor's width rather than two independent findings.

Full anchor-sweep permutation calibration yields $p = 0.0753$ (†). This is a *maximum-over-anchors* p-value: it accounts for the search but does not also correct for the parallel spacing sweep (Stage 1) or for the iterative keyword refinement of the dome sub-corpus (Section 3.3). Read against the conventional 0.05 threshold for unconditional inference, the result is marginal rather than confirmatory. The selected (ω, ϕ) frame is therefore used as an *exploratory representation* for conditional diagnostics in Stages 3–7, not as a globally significant discovery from the parameter search.

Candidate anchors outside the corridor do not show comparable enrichment under the same sweep: Mecca ($p = 0.2825$, ns), Mt. Meru ($p = 0.3956$, ns), Mt. Kailash ($p = 0.3956$, ns). These comparisons are reported because they will reappear in Section 5.2 where they constrain the mechanism behind the within-frame pattern.

4.3 Stage 3: Tier Thresholds

Sites are assigned to descriptive tiers by their deviation δ from the nearest harmonic node:

$$\delta_i = \left|(\lambda_i - \phi) \bmod \omega\right|, \quad \text{folded to } [0,\, \omega/2].$$

Tier-A++ ($\delta \leq 0.06°$, $\lesssim 6.66$ km; null rate 4%); Tier-A+ ($\delta \leq 0.15°$, $\lesssim 16.6$ km; null rate 10%); Tier-A ($\delta \leq 0.3°$, $\lesssim 33$ km; null rate 20%). Kilometer labels are equatorial approximations; no latitude correction is applied. Tier widths are derived from the von Mises dispersion of the Wikidata stupa corpus ($\hat{\sigma} = 0.1922°$, $\hat{\kappa} = 6.17$) so that they do not depend on the UNESCO data (Appendix B). As noted in Section 3, this means the Wikidata corpus is partly in-sample for the Stage 7 check; OWTRAD is out-of-sample with respect to parameter derivation, though it remains within the same Eurasian footprint.

4.4 Stage 4: Within-Frame Diagnostics

All results in this subsection are conditional on the selected (ω, ϕ) frame. Holm step-down adjustment is applied within the primary family ($k = 3$). The p-values do not generalise to the unconditional question (Stage 5).

Empirical observation. The full corpus has a Tier-A+ rate of 13.1% (132/1011; Clopper-Pearson 95% CI [11.0%, 15.3%]) against the 10% geometric null (Figure 1). The within-frame enrichment is 1.31× the null rate. This is the weak selected-frame signal audited in the remainder of the paper.

Test 1 (dome von Mises mixture, exploratory subset). A two-component von Mises/uniform mixture fit to the dome sub-corpus ($N = 90$) yields $\hat{w} = 0.0876$ ($\approx 8.8\%$ of dome sites concentrated at the selected phase; $\hat{\kappa} = 252.19$), $p_{\text{boot}} = 0.00180$ (**; Holm $p_{\text{adj}} = 0.004$, **). Because the dome keyword set was refined iteratively during analysis, this test is exploratory subset characterisation; the Holm adjustment within the primary family does not absorb the keyword-selection step. We report it as descriptive of the iteratively-refined sub-corpus, not as confirmatory inference.

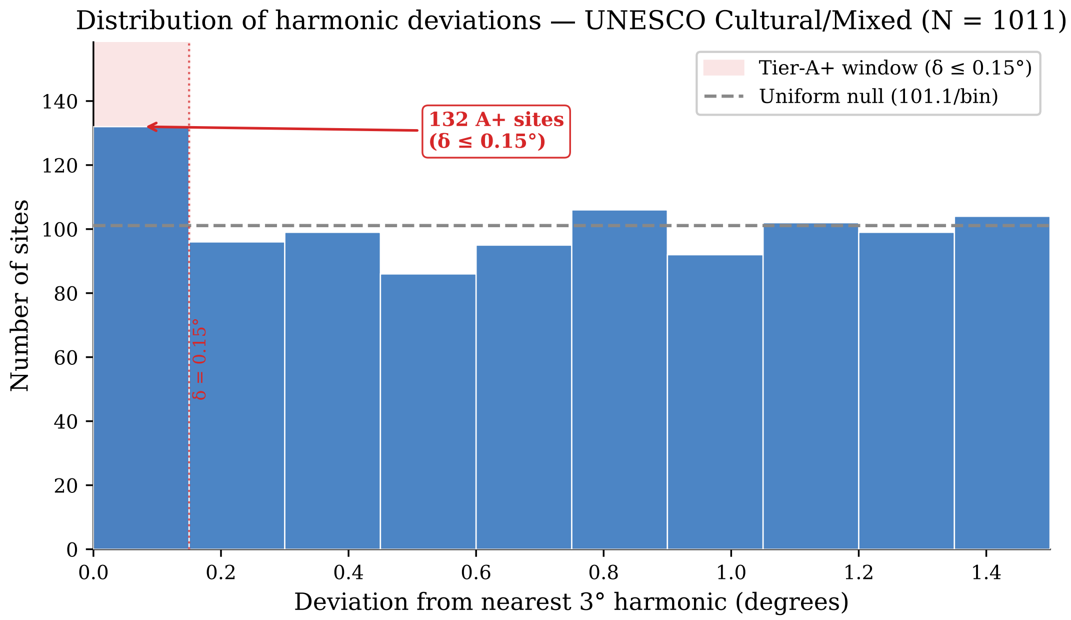

Figure 1: Harmonic deviation distribution for the full UNESCO corpus ($N = 1011$) under the selected 3° frame. The shaded region marks the Tier-A+ window ($\delta \leq 0.15°$); the dashed line marks the 10% geometric null. The first-bin excess is the within-frame structure that Stages 5–7 evaluate. As shown in Section 4.5, the corresponding invariant test on raw longitudes is null.

Table 1: Stage 4 diagnostics: within-frame tests conditional on the selected 3° / ∼35°E frame. All p-values are conditional on frame selection and do not address the unconditional question evaluated at Stage 5 (Section 4.5).

Test	N	Raw p	Holm p_{adj}	Role
Test 1 (dome VM mix.)	90	0.00180[b]	0.004 **	Exploratory subset
Test 2 (Spearman ρ)	1011	0.00065[b]	0.002 **	Within-frame diagnostic
Test 3 (mean dev.)	1011	0.04729[b]	0.047 *	Within-frame diagnostic

[b]permutation or bootstrap p. Holm (1979) step-down (Holm, 1979). Holm correction is applied within the conditional family; it does not absorb the Stage 1 spacing sweep, the Stage 2 anchor sweep, or the iteratively-refined keyword set used in Test 1.

Test 2 (Spearman cluster asymmetry). $\rho = -0.1076$, permutation $p = 0.00065$ (***; Holm $p_{\text{adj}} = 0.002$, **). Nodes drawing more sites show smaller mean deviations. This is the expected signature when a scoring function with a smooth attractor is applied to clustered data: density modes pull the nearest grid node toward the modal centre, reducing the mean deviation of the cluster's members. The result is informative about the data–scoring interaction; it is not, on its own, evidence of a historical mechanism.

Test 3 (phase-rotation null). The selected reference phase minimises the full-corpus mean harmonic deviation relative to random grid placements ($p_{\text{perm}} = 0.04729$, *; Holm $p_{\text{adj}} = 0.047$, *). This is the within-frame analogue of the Stage 2 anchor-specificity finding.

Dome sub-population detail (illustrative). As a descriptive result, 18/90 dome sites fall at Tier-A+ (20.0%, 2.00×; Fisher $p = 0.0346$, *). Tier-A++ concentration is stronger: 11/90 (12.2%, 3.06× the 4% null; permutation $p = 0.004$, **; bootstrap $p = 0.006$, **). These figures are reported for transparency; their interpretation is bounded by the iterative keyword construction of the dome sub-corpus.

Leave-n-out sensitivity. No single A+ site drives the dome result: removing any A+ site yields $N = 89$, A+= 17, $p = 0.0067$. Leave-3-out worst case across all $\binom{18}{3}$ triples: $p = 0.0253$. The A++ sub-population: leave-3-out worst case $p = 0.0233$ (leave-2-out: $p = 0.0087$). Robustness to specific sites is not, however, robustness to the keyword construction of the sub-corpus.

4.5 Stage 5: Boundary-Condition Tests — The Main Limitation

The Stage 4 results are conditional on a selected frame chosen from a parameter sweep. The broader question — whether the data support 3° periodicity without reference to any particular phase — is the question Stage 5 addresses.

The full-corpus Rayleigh test for 3° periodicity returns $R = 0.0224$ ($Z = 1.06$, $p_{\text{perm}} = 0.3716$, ns). This test does not support invariant 3° periodicity in the raw longitude distribution.

This is the main limit on interpretation. Disagreement between Stage 4 (conditional, positive) and Stage 5 (invariant, null) localises the signal at the selected-frame level rather than validating it as invariant periodicity. A user who reads the Stage 4 p-values without the Stage 5 result would draw a stronger inference than the data support. The audit is structured so that the Rayleigh null remains a binding boundary condition on the conditional results.

A ±0.3% shift from the selected 3° spacing collapses joint within-frame concentration across the dome sub-population. This underscores the frame-dependence of the signal; it does not by itself settle the choice among alternative explanations.

4.6 Stage 6: Geographic-Concentration Nulls

Domed monuments are concentrated in the Eurasian corridor (71% in 20° to 120°E versus 41% of the full corpus). Two null models ask whether this footprint, rather than any morphological-spatial relationship, accounts for the within-frame enrichment of the dome sub-population. Because the keyword set was iterative, these results are treated as exploratory diagnostics.

Null A (within-dome bootstrap). Drawing $N = 90$ longitudes from the dome sites' own longitude list yields a mean expected A+ count consistent with the observed value

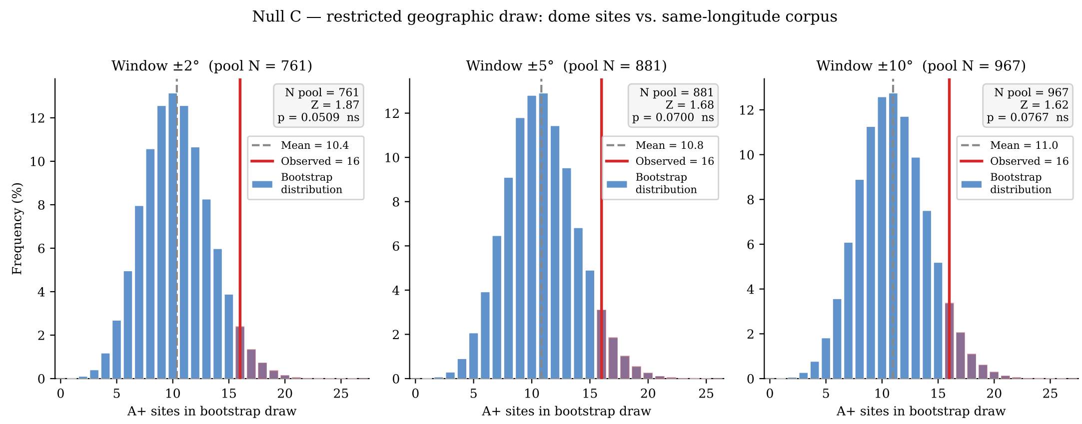

Figure 2: Null C bootstrap distributions for the context-validated dome sub-population ($N = 83$, observed A+ $= 16$) at three window widths. Null C is consistent with morphological specificity at the nominal level but does not survive correction for the three-window family.

($p = 0.5429$, ns). The dome enrichment is consistent with where dome sites already are; shared geography is therefore a sufficient alternative explanation for the subset pattern.

Null C (restricted geographic draw). Drawing from all full-corpus sites within $\pm W°$ of any dome longitude, the dome subset nominally exceeds the broader footprint at all three tested windows: $\pm 2°$ ($p = 0.0257$, *); $\pm 5°$ ($p = 0.0393$, *); $\pm 10°$ ($p = 0.0442$, *). These marginal results do not survive Bonferroni correction across the three windows ($\alpha_{\text{adj}} = 0.0167$).

The two nulls return ambiguous evidence on whether any morphological specificity remains after geographic concentration is accounted for: Null A is consistent with footprint alone being sufficient; Null C is nominally consistent with residual morphological specificity but does not survive correction. The pipeline reports both rather than forcing a resolution; ambiguity at this stage is what the diagnostic is designed to surface.

Out-of-footprint support. The UNESCO Americas sub-corpus ($N = 145$) shows an A+ rate of 9.7%, below the 10% null ($p = 0.5149$, ns, one-sided depletion). This small sub-corpus does not support extension of the selected-frame pattern outside the Eurasian footprint. It is not large enough to rule out a signal elsewhere, but it gives no positive out-of-footprint validation.

5 Cross-Corpus Stability under Frozen Parameters: A Same-Footprint Check

Stage 7 asks whether the selected frame organises independently assembled corpora at frozen parameters. Read in isolation, the answer is yes: both stability-check corpora show elevated enrichment under the same (ω, ϕ) that selected the within-frame pattern in UNESCO. Read in combination with Stages 5 and 6, the apparent stability does not warrant an invariant interpretation. This section documents both the empirical stability and the limits on what it can mean.

5.1 Stability Under Frozen Parameters

Wikidata Q180987 stupas. Under the identical anchor, spacing, and tier thresholds, $N = 229$ stupas yield 70/229 Tier-A sites (30.6%, 1.53× null; binomial $p < 0.001$, ***; permutation $p_{\text{perm}} = 0.065$, $\sim$). Harmonic nodes with at least one Tier-A stupa carry 2.56× more stupa sites on average than empty nodes. The Java/Sumatra cluster (105° to 115°E) concentrates 42/54 sites at Tier-A (77.8%; OR = 18.38×, $p < 0.001$, ***). The Wikidata result is partly in-sample because the corpus was used in Stage 3; the figures are presented as a consistency check on the tier-derivation choice and as same-footprint stability, not as independent replication.

OWTRAD Silk Road network. Testing all 3892 endpoint positions across the full edge list (each city weighted by the number of segments it connects), 211 fall at A++ harmonics, 457 at A+, and 886 at A (1.36× null; $z = 4.53$, $p < 0.0001$, *** for A++; $p = 0.0002$, *** for A+). A phase-shift test (random offset of the 3° grid, 100000 iterations) gives $p = 0.0202$, *, consistent with conditional enrichment at the selected phase. Harmonics containing at least one A++ endpoint (30 harmonics) carry 2.70× more route connectivity than harmonics without one (11 harmonics; $p = 0.0007$, ***, Mann-Whitney one-tailed). OWTRAD is out-of-sample with respect to the tier-derivation and UNESCO keyword steps, but it shares the same Eurasian footprint.

5.2 Why the Stability Is Not Independent Replication

A natural reading of the Stage 7 results is: "three corpora, assembled for different purposes, show the same structure under identical parameters — this must reflect something invariant." The audit asks what would be needed for that reading to become confirmatory. Three constraints prevent that inference here.

Constraint 1: The invariant test is null. The Rayleigh test on raw longitudes returns $p_{\text{perm}} = 0.3716$. The cross-corpus stability is observed under the selected frame; it does not validate as an invariant-period signal.

Constraint 2: Out-of-footprint support is lacking. The Americas sub-corpus gives $p = 0.5149$ for one-sided depletion. This does not rule out a signal elsewhere, but it does not support extension beyond the footprint shared by the three positive corpora.

Constraint 3: All three corpora share footprint. The Wikidata stupa corpus and the OWTRAD route network both sit in the same Eurasian corridor as the bulk of UNESCO Cultural inscriptions (Section 3). Under that condition, any frame that places

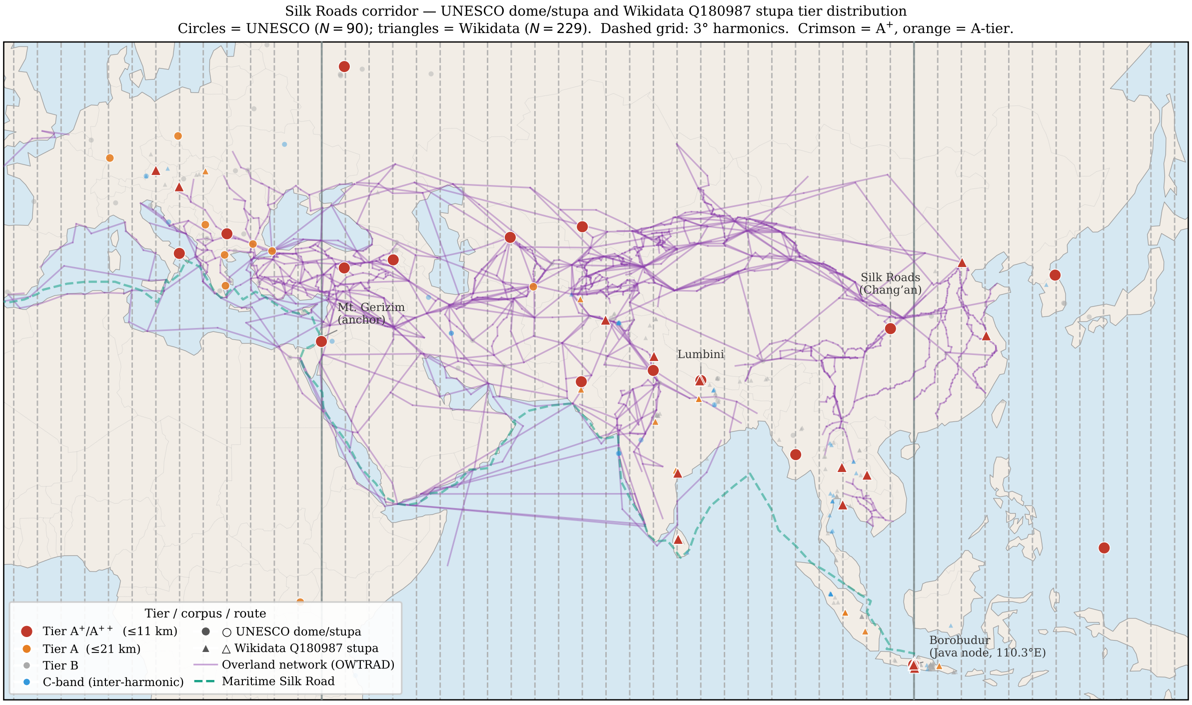

Figure 3: Tier distribution of UNESCO dome/stupa and Wikidata Q180987 stupa sites along the Eurasian corridor, overlaid on OWTRAD attested trade routes (Ciolek, 2004). The 3° harmonic grid (dashed) is the frame selected at Stage 2. The figure illustrates what cross-corpus organisation under frozen parameters *looks like* when the test corpora share geographic footprint with the primary corpus. Section 5.2 discusses why this visual coherence does not, on its own, license an invariant interpretation.

lattice nodes near dense regions of the corridor may produce elevated enrichment in any of the three corpora. Cross-corpus agreement of this kind is same-footprint stability under a single representational choice on three views of a common spatial concentration, not independent replication of an invariant phenomenon.

Anchor-specificity check. A reader might object that not every anchor produces the observed enrichment, citing the Stage 2 finding that Mecca ($p = 0.2825$), Mt. Meru ($p = 0.3956$), and Mt. Kailash ($p = 0.3956$) all return non-significant conditional p-values in UNESCO under the 3° frame. This is correct, and it is one reason the selected frame is an empirical object worth auditing. It does not, however, bear the inferential weight a strong reading would require. The 35°E corridor was selected by the maximum-over-anchors search because, given the 3° spacing, it places lattice nodes near dense regions of the Eurasian corridor. That corridor specificity is consistent with a geographic-fitting mechanism as a sufficient alternative explanation.

5.3 Scope of the Stability Finding

All three corpora are concentrated in the Eurasian corridor. The worked example therefore cannot, on its own data, distinguish two explanations of the cross-corpus stability:

1. the selected frame captures something specific to the 3° spacing and ∼35°E phase that goes beyond Eurasian corridor geography;
2. the three corpora share enough geographic structure that any frame fitting one will fit the others, and shared geography is sufficient.

The Stage 5 invariant null, the Stage 6 Americas result, and the partial in-sample status of the Wikidata check prevent (1) from becoming confirmatory. The honest summary is therefore: the worked example contains a weak selected-frame signal, but the present tests do not warrant an invariant interpretation; shared geography plus frame selection is a sufficient alternative explanation for the observed same-frame behaviour.

What would change the conclusion. Two complementary forms of evidence would be required to revisit (1). First, a heritage corpus assembled outside the Eurasian corridor — for example, national registers from sub-Saharan Africa, the Pacific, or Mesoamerica — analysed at frozen parameters by an independent team using the deposited code, with elevated enrichment that mirrored the in-corridor result. Second, an invariant test (Stage 5) that returned positive on the combined corpus, supporting periodicity rather than phase-fit at a particular anchor. Neither is available in the present data; we identify both as the priority follow-up tests, and neither has been carried out.

A note on pre-registration. Conventional pre-registration infrastructure (OSF, AsPredicted) is well-suited to controlled experimental designs and less well-suited to single-investigator spatial-analysis pipelines that depend on iterative parameter characterisation. The present work substitutes a frozen Zenodo deposit (DOI: 10.5281/zenodo.19938283, v1.6.0) of the analysis code and selected parameters. Independent application of the deposited pipeline to corpora outside the Eurasian footprint, by a different team, is the pre-registration analogue this design supports. The substitution is imperfect (the deposit is contemporaneous with the present analysis, not strictly antecedent), and it should be read as such.

6 Discussion

6.1 What the Pipeline Distinguishes

The worked example separates three claims that prior studies of spatial periodicity have tended to conflate.

Within-frame concentration. The selected frame organises the UNESCO corpus and two related datasets in a way that random permutations of the same frame do not (Tests 1–3, all conditional on the selected frame, surviving Holm correction within the conditional family). This is a genuine property of the data *under the chosen representation*.

No validation as invariant periodicity. The full-corpus Rayleigh test is null ($p_{\mathrm{perm}} = 0.3716$). The within-frame concentration does not extend to an invariant 3° period in raw longitude. The pipeline distinguishes these two statuses by design, and the second status limits interpretation.

Same-footprint cross-corpus stability. The same frame produces elevated enrichment in three Eurasian corpora at frozen parameters but not at other reference longitudes. This is interesting under the chosen representation, but it is consistent with a shared-geography account and is not separable from that account on the present data.

6.2 Mechanism

The within-frame concentration can be explained by the interaction of geographic clustering, the harmonic-deviation scoring function, and frame selection. The UNESCO corpus is clustered along the Eurasian corridor; the parameter sweep selects a frame that places lattice nodes near dense regions of that clustering; the scoring function rewards proximity to those nodes. This interaction is sufficient as an alternative explanation for stable within-frame structure without any stronger historical mechanism.

The same explanation also accounts for why cross-corpus stability appears when test corpora share footprint with the primary corpus. The Americas result does not support extension outside that footprint. The paper therefore leaves the signal unresolved rather than assigning it to a stronger historical mechanism.

6.3 Comparison to Prior Approaches

The pipeline addresses specific weaknesses of prior work. Thom (1967) and Kendall (1974) used investigator-assembled site lists; this pipeline uses externally curated corpora. Freeman (1976) criticised ad hoc unit choices; this pipeline sweeps over candidate units and reports the selection. Prior studies rarely distinguished frame-dependent structure from invariant periodicity; the pipeline's Stage 5 makes that distinction explicit and visible. Prior studies also did not implement cross-corpus stability checks at frozen parameters; the present worked example shows why such checks, when they are not paired with invariant tests and out-of-footprint validation, can be overinterpreted.

The general issue — analyses that survive every check the analyst chose to run, but reflect researcher degrees of freedom rather than the underlying phenomenon — is articulated for the broader social and biomedical literature by Gelman & Loken (2014).

The present contribution applies that diagnostic stance to the specific structure of spatial heritage analysis.

7 Implications for Practice

The worked example yields four practical recommendations for spatial analysis of cultural heritage data.

Report the search. State how many frames, phases, spacings, anchors, or thresholds were considered, and at what resolution. Distinguish frame selection from subsequent diagnostics. When the same data select the frame and test it, the resulting p-values are conditional on the selected frame and should be labelled as such. Holm or Bonferroni adjustment within a conditional family does not absorb the prior selection step; that step requires its own treatment (maximum-over-search permutation, or hold-out).

Run an invariant test, and report its limit. A coordinate-aligned result should always be checked against at least one method that does not assume the selected reference frame. The Rayleigh test on raw longitudes is the canonical choice for periodicity claims. When the invariant test is null and the within-frame test is positive, the honest summary is that the result remains selected-frame structure under the present tests. The within-frame p-values do not generalise across that boundary.

Distinguish same-footprint from out-of-footprint cross-corpus checks. When multiple corpora are available, freeze the selected frame and apply it to each — but report which corpora share geographic footprint with the primary corpus and which do not. Same-footprint agreement is shared-representation stability, not independent replication. Out-of-footprint corpora are the genuine validation check. If only same-footprint corpora are available (as in the present worked example), the stability stage should be labelled as a within-domain consistency check, not as cross-corpus replication.

Audit corpus structure. Geographic clustering, corpus overlap, partial in-sample use of stability-check corpora (e.g., through tier derivation), and iterative keyword selection should all be treated as part of the inferential design. When multiple corpora share geographic structure, cross-corpus consistency of frame-fitted results may be expected; it should not carry the inferential weight that "independent replication" suggests.

8 Conclusion

This paper contributes a reproducible audit pipeline for evaluating frame-dependent structure in spatially clustered heritage data. The pipeline's seven stages — unit discovery, anchor characterisation, tier-threshold derivation, within-frame diagnostics, boundary-condition tests, geographic-concentration nulls, and cross-corpus stability checks — are demonstrated on the UNESCO Cultural and Mixed corpus with stability checks against Wikidata Q180987 stupas (within-corridor, partly in-sample) and the OWTRAD Silk Road network (within-corridor, out-of-sample with respect to parameter derivation).

The paper does not claim invariant 3° periodicity. It also does not claim that the observed selected-frame signal is meaningless. The worked example contains a weak within-frame signal: a parameter sweep selects a frame near 3° spacing and the ∼35°E corridor; the UNESCO corpus shows 132/1011 Tier-A+ sites (13.1% versus the 10% geometric null); Stage 4 conditional diagnostics are positive; and Wikidata and OWTRAD show related same-frame behaviour under frozen parameters. Those facts locate a real selected-frame pattern for audit.

The pattern is not strong enough, independent enough, or invariant enough to support a substantive claim of spatial periodicity. The full-corpus Rayleigh test is null ($p_{\text{perm}} = 0.3716$), the Americas sub-corpus does not support extension outside the Eurasian footprint, the dome subset is exploratory because the keyword filter was iterative, and the positive cross-corpus checks share the same broad geography as the primary corpus. Shared geography plus frame selection is therefore a sufficient alternative explanation for the observed stability.

The correct status of the result is weak, frame-dependent, geographically confounded, and not validated as invariant. Future validation would require preregistered or otherwise frozen-parameter tests on genuinely out-of-footprint heritage corpora, ideally by an independent team using the deposited code, together with a positive invariant test. Until such evidence exists, the 3° / ∼35°E result should be treated as a non-confirmatory worked example of how weak selected-frame signals can arise and how they can be audited without being overinterpreted. All code, data, and reproduction instructions are archived at Zenodo (DOI: `https://doi.org/10.5281/zenodo.19938283`, v1.6.0).

Data and code availability. All analysis scripts, data files, and reproduction instructions are archived at Zenodo (DOI: `https://doi.org/10.5281/zenodo.19938283`, v1.6.0). The pipeline regenerates all statistics and LaTeX macros from raw data without manual editing (`bash manuscript/reproduce_all_macros.sh`). Full audit files, keyword lists, and supplementary figures are provided in Supplementary S1.

A Tier-Threshold Sensitivity

A log-scale null-rate sweep across 73 thresholds shows that the dome within-frame pattern is not limited to the specific tier width: 57 of 73 thresholds reach $p < 0.05$ for the dome subset; dome OR ranges 1.53×–3.40× across the three canonical tiers. Within ±1 log-decade of the A+ threshold, 27 of 28 thresholds meet the nominal $p < 0.05$ threshold. As elsewhere, these results are conditional on the selected (ω, ϕ) frame and do not generalise to the unconditional question (Section 4.5).

B Von Mises Mixture Characterisation

A von Mises + uniform mixture fitted to the harmonic phase deviations of the stability-check corpora under the selected frame provides within-frame dispersion estimates used for tier-threshold derivation (Stage 3): Wikidata Q180987 stupas ($N = 229$): $\hat{\kappa} = 6.17$, $\hat{w} = 0.1297$, $\hat{\sigma} = 0.1922°$. OWTRAD route nodes ($N = 1674$): $\hat{\kappa} = 9.58$, $\hat{w} = 0.0229$, $\hat{\sigma} = 0.1543°$. N-weighted pooled $\hat{\sigma} = 0.1588°$; κ ratio $= 1.6$ across corpora. Tier boundaries correspond to $0.25\hat{\sigma}$, $0.50\hat{\sigma}$, and $1.00\hat{\sigma}$ of the stupa corpus dispersion.

C Keyword Audit

7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) matched as whole words against site names, XML descriptions, and extended OUV text. Context-validation for ambiguous terms: Gate 1 (28 negative patterns) rejects non-architectural uses; Gate 2 (86 positive patterns) requires architectural confirmation. Seven sites rejected in the context-validated subset: Hiroshima Peace Memorial, Richtersveld, Gyeongju, Viñales Valley, Rock Islands Southern Lagoon, Uluru–Kata Tjuṭa, and Diquís Stone Spheres. The keyword set was refined iteratively during analysis; results from the dome sub-corpus are accordingly hypothesis-generating, as flagged in Sections 3.3 and 4.4. Full site-by-site audit in the Zenodo archive (Tenenbaum, 2026).

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